Mahfujur Rahaman Niyal

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Personal Statement

Dedicated Computer Science and Engineering student at AIUB with interests in Software Requirements Engineering, Software Quality Assurance and Testing, Python, Network Security, and Management Information Systems. Experienced in software development using Python, Java, and SQL, with teamwork, leadership, and event coordination experience. Committed to continuous learning and developing practical technology solutions.

Educational Background

B.Sc. in Computer Science and Engineering Jan. 2023 – 2027

American International University-Bangladesh (AIUB)
Major: Information Systems | Minor: Software Engineering

Higher Secondary Certificate – Science Jul. 2019 – Dec. 2021

Dularhat Adarsha College, Bhola
GPA: 5.00 / 5.00

Secondary School Certificate – Science Jan. 2017 – Mar. 2019

Dularhat Secondary School, Bhola
GPA: 4.00 / 5.00

Projects

UniRent – Software Requirements Specification (SRS) Documentation Software Engineering

Prepared a comprehensive SRS document for a student rental web platform featuring identity verification and deposit protection. Reviewed by MD Forhadul Islam, Product Manager at Innospace Infotech Ltd.

Online Jersey Shop Management System Java

Developed a Java-based project for an online jersey shop.

Tic-Tac-Toe Game HTML, CSS, JavaScript

Created a browser-based game using HTML, CSS, and JavaScript.

Smart Push Rehab – Sensor-Based Tangible User Interface for Stroke Rehabilitation HCI & IoT

Prepared project documentation and system design for a hybrid physical-digital rehabilitation tool featuring force/angle tracking, ESP32 real-time feedback, and a web dashboard under the guidance of Prof. Khandaker Tabin Hasan, PhD.

NiyalGPT – AI Chatbot Full-Stack Web Development

Developed a full-stack AI chatbot using Gemini API with chat history, custom assistant personalities, and Markdown/code rendering.

Snake Game JavaScript

Developed a browser-based Snake game using JavaScript.

Travel Guide Project PHP

Created a PHP-based project related to travel information.

Client Portfolio Website HTML

Created a portfolio website using HTML.

Smart Local Service Management for Trusted Home Solutions Software Engineering

Managed project backlogs, user stories, and test cases using Trello for a home service management system covering admin, employee, and customer workflows.

AI-Based Study Planner Web Development

Developed a study planning web application with user authentication, dashboard, analytics, and settings features.

Technical Skills

Programming Languages: C, C++, C#, Java, Python, JavaScript, PHP, SQL, HTML, CSS

Computer Skills: MS Word, MS Excel, MS PowerPoint, Photoshop, Google Workspace, Canva, WordPress

Technical Tools: GitHub, Visual Studio, SQL Server, Oracle, Code::Blocks, Notepad++, MATLAB, Jira, Trello. Selenium, Kali Linux, Cisco Packet Tracer, Wireshark, AutoCAD

Soft Skills: Time Management, Problem Solving, Event Coordination, Communication, Teamwork, Customer Management, Decision-Making, Leadership, Ethics

Research

HeartGuard-XM: A Multimodal Explainable Framework for Coronary Artery Disease Assessment Fusing X-Ray Angiography with Interpretable Clinical Models – Thesis

Jul. 2026 – Present

An Explainable Artificial Intelligence Framework for Financial Fraud Detection with Risk-Aware Prediction and Interactive Decision Support

May 2026

Social Media-Based Cyberbullying and Its Effects on Teenagers in Bangladesh: A Quantitative Study with Mobile-Based Solution Proposal

Apr. 2025

Experience

Private Tutor (Classes 1–10) Jun. 2022 – Present

- Teach Physics, Chemistry, and Mathematics using simple explanations tailored to students' learning needs.
- Prepare lessons and practice materials to improve students' understanding and academic performance.
- Adapt teaching methods based on individual progress and learning difficulties.

Volunteer Jan. 2019

- Assisted with event preparation, logistics, and participant coordination.
- Supported on-site operations during education events and competitions.

Courses and Training

IT Essentials – Cisco Networking Academy Issued Feb. 2023

Extracurricular Activities

Personal Development and Technical Activities

- Collaborative development and launch of a live website.
- Self-learning and exploration of web development technologies.
- Practicing Git and GitHub for version control.

Participation and Certificates

- **Artificial Intelligence & Machine Learning Fundamentals Certificate** – Grameenphone Academy, 2026.
- **Think AI Certificate** – Grameenphone Academy, 2026.
- **Microsoft Excel Certificate of Completion** – OfficeMaster Academy, 2026.
- **Introduction to IoT and Digital Transformation Certificate** – Cisco Networking Academy & Grameenphone Academy, 2026.
- **AI-Pro Aptitude Hacks Certificate** – Grameenphone Academy, 2026.
- **Network Smarter with AI Certificate** – Grameenphone Academy, 2026.
- **Research Presentation Certificate** – 7th IEOM 2024 Conference, AIUB, 2024.
  Impact of Social Media and Reels on Concentration and Behavioral Changes in Youth Generation: Insights from Bangladesh
- **IT Essentials Certificate** – Cisco Networking Academy, 2023.
- **Participant, Workshop on Seminar on Software Requirement Engineering: Industry Practices** – AIUB Computer Club (ACC) & AIUB, 2026.
  Resource Person: Md. Shajahan Islam Sani, Project Manager II, Brain Station-23 PLC.